WORKED SOLUTION

STEP III 2007, Question 10

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q10

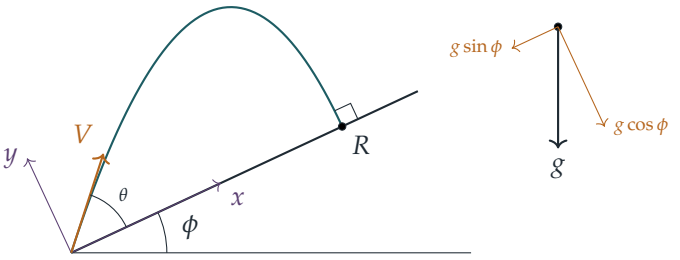
The key idea

Work in axes along and perpendicular to the slope, so that gravity becomes two constant accelerations, $g\sin\phi$ down the slope and $g\cos\phi$ into it. The striking condition is that the particle retraces its path. A smooth plane can only push perpendicular to itself, so it can reverse the velocity only if the particle arrives travelling straight into the plane. That single fact gives both results; the rest is algebra and an AM–GM minimisation.

Solution

The equations of motion

We model the particle as a point mass moving under gravity alone, with no air resistance. The weight mg acts vertically downwards. Resolving along the slope and perpendicular to it:



the acceleration is $-g\sin\phi$ in the x-direction and $-g\cos\phi$ in the y-direction, both constant. Integrating twice with the given initial velocity and starting point $(0,0)$,

$$x = Vt\cos\theta - \tfrac{1}{2}gt^2\sin\phi, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2\cos\phi.$$

The condition for retracing the path

The particle first returns to the plane when $y = 0$ with $t > 0$, that is at

$$t = T = \frac{2V \sin\theta}{g \cos\phi}.$$

Ask yourself What does "returns along the same path" tell us about the impact?

After the bounce, the particle again moves under constant gravity. If it is to run back along the same parabola to the origin, its velocity just after the bounce must be exactly the reverse of its velocity just before. (Think of running the film backwards.) We take the plane to be smooth, so the impulse it gives the particle is perpendicular to the plane: the component $\dot{x}$ along the slope is unchanged by the bounce. That component can only be reversed if it is zero. So the particle must strike the plane at right angles, with $\dot{x} = 0$ at $t = T$. (We also need the bounce to reverse $\dot{y}$ without loss of speed, so the impact must be perfectly elastic. The question takes this for granted.)

Now $\dot{x} = V\cos\theta - gt\sin\phi$, so $\dot{x}(T) = 0$ gives

$$V\cos\theta = gT\sin\phi = \frac{2V\sin\theta\sin\phi}{\cos\phi}.$$

Dividing by $V\cos\theta$, which is positive since $0 < \theta < \pi/2$,

$$2\tan\phi\tan\theta = 1,$$

as required.

The range

From $\dot{x}(T) = 0$ we have the neater form $T = \dfrac{V\cos\theta}{g\sin\phi}$. Then

$$\begin{aligned} R = x(T) &= V\cos\theta \cdot \frac{V\cos\theta}{g\sin\phi} - \frac{1}{2}g\sin\phi \cdot \frac{V^2\cos^2\theta}{g^2\sin^2\phi} \\ &= \frac{V^2\cos^2\theta}{g\sin\phi} - \frac{V^2\cos^2\theta}{2g\sin\phi} = \frac{V^2\cos^2\theta}{2g\sin\phi}, \end{aligned}$$

which is the required result. (Equivalently: x is greatest when $\dot{x} = 0$, so the landing point is the highest point up the slope that the motion would reach.)

The identity for $2V^2/(gR)$

From the range,

$$\frac{2V^2}{gR} = \frac{4\sin\phi}{\cos^2\theta} = 4\sin\phi\,(1 + \tan^2\theta).$$

Using $\tan\theta = \dfrac{1}{2\tan\phi}$, so $\tan^2\theta = \dfrac{\cos^2\phi}{4\sin^2\phi}$,

$$\frac{2V^2}{gR} = 4\sin\phi + \frac{\cos^2\phi}{\sin\phi} = 4\sin\phi + \frac{1-\sin^2\phi}{\sin\phi} = 3\sin\phi + \operatorname{cosec}\phi.$$

The largest range

For a given V, R is largest when $3\sin\phi + \operatorname{cosec}\phi$ is smallest. Write $s = \sin\phi$; since $0 < \phi < \pi/2$, $0 < s < 1$. By the AM–GM inequality for the positive numbers $3s$ and $1/s$,

$$3s + \frac{1}{s} \geq 2\sqrt{3s \cdot \frac{1}{s}} = 2\sqrt{3},$$

with equality exactly when $3s = 1/s$, that is $s = 1/\sqrt{3}$. This value does lie in $(0,1)$, so the bound is attained by a genuine slope. Hence

$$\frac{2V^2}{gR} \geq 2\sqrt{3} \quad\Longrightarrow\quad R \leq \frac{V^2}{\sqrt{3}\,g},$$

with equality when $\sin\phi = 1/\sqrt{3}$. The largest possible range is therefore $V^2/(\sqrt{3}\,g)$.

Ask yourself Is the condition $\phi + \theta < \pi/2$ satisfied at the maximum?

Yes, and in fact always. With $\tan\theta\tan\phi = \frac{1}{2}$ and both angles acute,

$$\tan(\phi+\theta) = \frac{\tan\phi + \tan\theta}{1 - \tan\phi\tan\theta} = 2(\tan\phi + \tan\theta) > 0,$$

and since $0 < \phi + \theta < \pi$, this forces $\phi + \theta < \pi/2$. At the maximum, $\sin\phi = 1/\sqrt{3}$ gives $\tan\phi = 1/\sqrt{2}$, so $\tan\theta = 1/\sqrt{2}$ too: the best case has $\theta = \phi \approx 35.3°$.

Checks

- Units: V^2/g is a length, as a range should be.
- Limits: as $\phi \to 0$, $\operatorname{cosec}\phi \to \infty$ and $R \to 0$. On a nearly flat plane, the only way to land at right angles is to fire almost straight up, which gives almost no range. As $\phi \to \pi/2$ the plane is a vertical wall, $\theta \to 0$, and $2V^2/(gR) \to 4$, so $R \to V^2/(2g)$: the height reached by a particle thrown straight up the wall. Both make sense.
- Numerically, with $g = 9.8$, $V = 10$ and $\phi = 0.7$, stepping the motion forward lands at $x \approx 5.8561$, matching the formula. The landing velocity has $\dot{x} \approx 0$, and reversing it brings the particle back to the origin. A grid search over ϕ gives maximum $R \approx 0.05891$ for $V = 1$, equal to $1/(9.8\sqrt{3})$.

Common slips

- Resolving gravity the wrong way round. The component *along* the slope is $g\sin\phi$ and the component *perpendicular* to it is $g\cos\phi$. Check with $\phi = 0$.
- Saying the particle must land "horizontally" or "at its highest point". The condition is that it hits the *plane* at right angles, $\dot{x} = 0$ in the tilted axes.
- Forgetting to check that $\sin\phi = 1/\sqrt{3}$ is attainable. AM–GM gives a bound, and it is only the maximum if equality can actually occur.

Beyond STEP

Retracing the path rests on the *time-reversibility* of Newtonian mechanics. For forces that depend only on position, reversing every velocity sends the system back along its history. Choosing axes adapted to the geometry, rather than the default horizontal and vertical, is a habit you will use constantly in university mechanics. Kleppner and Kolenkow's *An Introduction to Mechanics* develops both ideas.

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